Ye Kyaw Thu

Language and Semantic Technology (LST),
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EDUCATION

D.Sc.	Global Information and Telecommunication Studies, Waseda University, Japan, 2011
M.Sc.	Global Information and Telecommunication Studies, Waseda University, Japan, 2006
B.Sc.	Physics, Dagon University, Myanmar, 2000
Diploma	One Year Special Japanese Language Course, The Japanese Language School of the International Student Institute, Japan, 2004
Diploma	International Advanced Diploma in Computer Studies, NCC Education, UK, 1999
Diploma	International Diploma in Computer Studies, NCC Education, UK, 1998

SCHOLARSHIP

3/2010-4/2011 Atsumi Scholarship, Japan

4/2006-3/2009 Japanese Government Scholarship (Monbukagakusho: MEXT)

4/2004-3/2006 Imaizumi Memorial Burmese Scholarship Foundation, Japan

1/2002-3/2002 Centre of the International Cooperation for Computerisation (CICC), Japan

APPOINTMENTS

1/2019–Present	National Electronics and Computer Technology Center (NECTEC), Pathumthani, Thailand Visiting Professor, Language and Semantic Technology (LST)
5/2019–Present	Language Understanding Lab., Pyin Oo Lwin, Myanmar Lab Leader
5/2022–2/2023	Cambodia Academy of Digital Technology (CADT), Phnom Penh, Cambodia Affiliate Professor, Institute of Digital Research & Innovation
12/2018–2/2021	Yangon Technological University (YTU), Yangon, Myanmar Visiting Professor
3/2017–2/2023	University of Technology (Yatanarpon Cyber City), Pyin Oo Lwin, Myanmar Visiting Professor
4/2012–3/2021	Department of Applied Mathematics, GITI, Waseda University, Tokyo, Japan Visiting Researcher
11/2018–10/2019	Bagan Innovation Technology, Yangon, Myanmar Technical Advisor
9/2016–3/2018	Okayama Prefectural University, Okayama Prefecture, Japan Researcher

4/2012–3/2016	Multilingual Translation Lab., Universal Communication Research Institute, NICT, Kyoto, Japan Researcher
4/2009–3/2012	Global Information and Telecommunication Studies (GITS), Waseda University, Tokyo, Japan Research Associate
1/2000–6/2002	Graviton Software Development Team, Yangon, Myanmar Founder/Lecturer
5/1998–11/1999	KMD Company Ltd., Yangon, Myanmar Project Manager

RESEARCH AREAS

In the realm of Natural Language Processing (NLP), my research is primarily focused on machine translation, automatic speech recognition (ASR), text-to-speech (TTS), and speech translation, with a particular emphasis on under-resourced languages such as Burmese and ethnic languages of Myanmar, as well as sign language and Braille systems. Within Human-Computer Interaction (HCI), my contributions center on identifying common text input methods and user interfaces suitable for Asian syllabic languages. In the domain of Artificial Intelligence (AI), I am engaged in advancing language acquisition for robots, aiming to enable seamless communication between humans and machines.

ACADEMIC SUPERVISION

I supervise graduate research students at several leading institutions, including Assumption University (AU), Kasetsart University, King Mongkut's Institute of Technology Ladkrabang (KMITL), Sirindhorn International Institute of Technology (SIIT), and the Japan Advanced Institute of Science and Technology (JAIST).

SELECTED PUBLICATIONS

Journal Articles

2025 Khaing Hsu Wai, **Ye Kyaw Thu**, Thazin Myint Oo, Nobuo Funabiki, Lu Xiqin, and Akihiro Yamamura, "A Study on AI-Driven Question Generation Using ChatGPT for Python Programming Education", Journal of Intelligent Informatics and Smart Technology, Vol 11, No.1, October 2025, pp. 1-7 (Submitted March 30, 2025; accepted July 16, 2025; revised August 25, 2025; published online October 31, 2025)

2024 Mya Ei San, Sasiporn Usanavasin, **Ye Kyaw Thu**, Manabu Okumura, "A Study for Enhancing Low-resource Thai-Myanmar-English Neural Machine Translation," ACM Transactions on Asian and Low-Resource Language Information Processing (TALLIP), doi.org/10.1145/3645111, (Submitted Aug 15, 2023; accepted with major revision Nov 4, 2023, accepted Jan 31, 2024.)

2023 Thazin Myint Oo, Thitipong Tanprasert, **Ye Kyaw Thu**, Thepchai Supnithi, "Transfer and Triangulation Pivot Translation Approaches for Burmese Dialects," in IEEE Access, vol. 11, pp. 6150-6168, 2023, doi: 10.1109/ACCESS.2023.3236804. (Received 15 October 2022, accepted 27 December 2022, date of publication 13 January 2023, date of current version 20 January 2023.)

2022 Zar Zar Hlaing , **Ye Kyaw Thu**, Thepchai Supnithi, Ponrudee Netisopakul, "Improving Neural Machine Translation with POS-Tag Features for Low-Resource Language Pairs,"" Heliyon Journal, doi.org/10.1016/j.heliyon.2022.e10375(Submitted 5 March 2022, Revised 25 June 2022, Accepted 15 August 2022, Published online 22 August 2022.)

2021 Zun Hlaing Moe, Thida San, Ei Thandar Phyu, Hlaing Myat Nwe, Hnin Aye Thant, Naw Naw, Htet Ne Oo, Thepchai Supnithi, **Ye Kyaw Thu**, "Myanmar Text (Burmese) and Braille (Mu Thit) Machine Translation Applying IBM Model 1 and 2", Journal of Intelligent Informatics and Smart Technology, April 1st Issue, 2021, pp. 18-26, https://jiist.aiat.or.th/assets/uploads/16195380940305LtWrJIIST-44-FinalVersion.pdf (Submitted February 8, 2021; accepted March 11, 2021; revised April 15, 2021; published online April 25, 2021.)

International Conference

2025 Khaing Hsu Wai, Hlaing Myat Nwe, Thura Aung, Kaung Khant Si Thu, Hsu Yee Mon, Seng Pan, Thiha Nyein, Yu Myat Moe, **Ye Kyaw Thu**, Thazin Myint Oo, "Myanmar Text Grade-Level Prediction using Statistical and Linguistics Features," In Proceedings of the 7th Workshop on Innovation Initiatives in NLP/AI/Education (IINAE 2025), Nov 12, 2025, Phuket, Thailand, pp. 1-6

2025 Shin Thant Phyo, Pyae Linn, Lynn Myat Bhone, Thet Hmue Khin, Eaint Kay Khaing Kyaw, **Ye Kyaw Thu**, "myFoodQA: A Multimodal Dataset for Evaluating Cultural and Visual Reasoning in Myanmar Gastronomy," In Proceedings of the 7th Workshop on Innovation Initiatives in NLP/AI/Education (IINAE 2025), Nov 12, Phuket, Thailand, pp. 1-6

2025 Ye Bhone Lin, Thura Aung, **Ye Kyaw Thu**, Thazin Myint Oo, "ASR Error Correction in Low-Resource Burmese with Alignment-Enhanced Transformers using Phonetic Features," In Proceedings of the 20th International Joint Symposium on Artificial Intelligence and Natural Language Processing (iSAI-NLP 2025), Nov 12 to 14, 2025, Phuket, Thailand, pp. 1-6

2025 Thura Aung, Eaint Kay Khaing Kyaw, **Ye Kyaw Thu**, Thazin Myint Oo, Thepchai Supnithi, "Enhancing Burmese News Classification with Kolmogorov-Arnold Network Head Fine-tuning," In Proceedings of the 20th International Joint Symposium on Artificial Intelligence and Natural Language Processing (iSAI-NLP 2025), Nov 12 to 14, 2025, Phuket, Thailand, pp. 1-6

2025 Kaung Lwin Thant, Kwankamol Nongpong, **Ye Kyaw Thu**, Thura Aung, Khaing Hsu Wai, Thazin Myint Oo , "myNER: Contextualized Burmese Named Entity Recognition with Bidirectional LSTM and fastText Embeddings via Joint Training with POS Tagging," the International Conference on Cybernetics and Innovations (ICCI 2025), April 2-4, Pattaya Chonburi, Thailand pp.1-6

Awards

2025 Our paper entitled "Enhancing Burmese News Classification with Kolmogorov-Arnold Network Head Fine-tuning" achieved **"the Best Paper Award"** at the the 20th International Joint Symposium on Artificial Intelligence and Natural Language Processing (iSAI-NLP 2025), Phuket, Thailand

2024 Our two-day project, titled "Real-time Dementia Awareness Monitoring Tool," with team members Igor Tatarnikov (UCL), Saowaluk Srikajornlarp (HPC5), and Ye Kyaw Thu (NECTEC), won the **"Innovation Champions Prize"** in the innovation challenge competition at the ThAiMed Initiative Workshop, held from August 26~28, 2024, in Bangkok, Thailand.

2024 Our paper entitled "Statistical vs Neural Machine Translations for Khmer Braille" achieved **"the 2nd Prize"** at the Workshop on Research and Innovation for Students 2024, Danang, Vietnam

2021 Our paper entitled "English to Burmese Machine Translation with Asian Pivot Languages" achieved **"the Best Paper Award"** and also one more paper entitled "Word to Word translation between Myanmar(Burmese) and other Ethnic languages" achieved "the Best Presentation Award" at the 3rd Joint Southeast Asian NLP/AI R&D Workshop, 2021

2021 Presenter Ms. Ni Htwe Aung achieved the **"Presentation Award"** for the paper entitled "CRNN Based OCR for American and British Sign Language Fingerspelling" at KSE2021, Thailand.

2020 Our workshop paper "Grapheme-to-IPA Phoneme Conversion for Burmese (myG2P Version 2.0)" achieved **"the Best Paper Award"** at the 2nd Joint Myanmar-Thai NLP/AI R&D Workshop, 2020, Bangkok, Thailand.

2020 Our conference paper "Statistical Machine Translation for Myanmar Language Paraphrase Generation" achieved **"the Best Paper Award"** at the Track 1: Natural Language Processing (NLP), iSAI-NLP 2020, Bangkok, Thailand.

2019 Our paper entitled "Statistical Machine Translation of Myanmar Dialects" got **"the Workshop Paper Award"** at the First Joint Myanmar-Thai NLP/AI R&D Workshop, 2019, Chiang Mai, Thailand

2015 **Best Paper Award for ICCA 2015**, "Syllable Pronunciation Features for Myanmar Grapheme to Phoneme Conversion," February 5~6, 2015. (NICT News, No.452 JUN 2015, Page No. 13)

2013 **Special Award (Emerging)** for SNLP 20013, "Unsupervised and Semi-supervised Myanmar Word Segmentation Approaches for Statistical Machine Translation," October 28~30, 2013. (NICT News)

2011 **Waseda University Grant for Special Research Projects** (Project number: 2011B-215), 2011

2009 **Waseda University Grant for Special Research Projects** (Project number: 2009A-901), 2009

2009 **3rd Prize for ACM Student Research Competition Grand Finals 2009**, "Positional Prediction: Common and User Friendly Text Input Interfaces for Asian Syllabic Languages on Mobile Devices", ACM Awards Banquet, San Diego, USA, June 27, 2009

2008 **Waseda University Expense Stipend, Student representative to the 9th APRU Doctoral Students Conference**, July 14~18, 2008, Vladivostok, Russia

2008 **Young Author Recognition Certificate** of ITU-T Kaleidoscope Academic Conference, "Positional Gesture for Advanced Smart Terminals: Simple Gesture Text Input for Syllabic Scripts Like Myanmar, Khmer and Bangla," May 12~13, 2008

2008 **2nd Prize for ACM Student Research Competition of CHI 2008**, "Positional Prediction: Consonant Cluster Prediction Text Entry Method for Burmese (Myanmar Language)," April 5~10, 2008

2007 **Best Paper Award for ACM MobileHCI 07**, "Positional Mapping Myanmar Text Input Scheme for Mobile Devices," September 9~12, 2007

1998 **1st Prize for student project of IDCS Diploma**, "Malaysia Airline Reservation System," 1998

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